

Windows Privilege Escalation - Enumeration Cheatsheet

Introduction

Enumeration is the first step in identifying potential attack vectors during privilege escalation. By systematically gathering details about users, services, files, and configurations, you can uncover misconfigurations or weaknesses in a Windows system. This guide organizes the essential enumeration techniques for easy reference.

User Enumeration

→Who Am I?

Use `whoami /all` to display current user details, including privileges and security groups.

→System Information

→`systeminfo`: Lists detailed system specs and patch levels.

→`hostname`: Reveals the machine's hostname, which might indicate its role in the network.

→Account Details

→`net users`: Lists all users on the system.

→`net user username`: Displays specific information about a given user.

→qwinsta: Identifies active user sessions on the system.

→Group Memberships

→net localgroup: Displays user groups defined on the system.

→net localgroup administrators: Identifies users in the Administrators group.

→Quick Restart (Optional)

Restarting may trigger startup processes:

Command	shutdown /r /t 0

Firewall Configuration

→Overview

Learn how the firewall is configured or bypassed. More details: [HackTricks Firewall Basics](#).

→Enable Remote Desktop (RDP)

To enable RDP access, execute these commands:

Command	reg add "HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Terminal Server" /v fDenyTSConnections /t REG_DWORD /d 0 /f
	netsh firewall add portopening TCP 3389 "Remote Desktop"

File and Directory Enumeration

→Finding Sensitive Files

→Search for patterns in text files:

Command	findstr /si password *.txt

→Use PowerShell to search for specific filenames across drives:
get-childitem c:\ -recurse -erroraction silentlycontinue |
where-object {\$_.Name -match "Veeam.one"}.

→Detailed File Information

Example: Check file details, including exact version: get-childitem
"C:\Program Files\Veeam\Veeam ONE\Veeam ONE
Agent\Veeam.One.Agent.Deployment.Service.exe" | format-list *.

Patch and Update Information

→Check Installed Updates

Command	wmic qfe get Caption,Description,HotFixID,InstalledOn
---------	---

Network Enumeration

→View Network Connections

→netstat -ano: Lists open connections with associated process IDs.

→Use PowerShell for additional details about process IDs: get-process -Id 4916.

Scheduled Tasks

→Identify and Manage Scheduled Tasks

→List tasks: schtasks.

→Detailed list: schtasks /query /fo LIST /v.

→Search tasks: schtasks | findstr /si soon.

→Delete tasks: schtasks /delete /tn soon /F.

Driver Information

→List Installed Drivers

Command	driverquery

Antivirus Status

→Check for Windows Defender

→sc query windefend: Verifies if Defender is present and running.

→wmic service get name,displayname,pathname,startmode | findstr "Defender".

→Disable Real-Time Monitoring

(Administrator session required): powershell -ep bypass -c Set-MpPreference -DisableRealtimeMonitoring \$true.

Service Enumeration

→List Running Services

Command	wmic service list brief findstr "Running"

Outro and Important Notes

Privilege escalation in Windows systems often relies on meticulous enumeration, as it reveals the weaknesses and misconfigurations that lead to potential escalation paths. Always approach these steps systematically, document findings thoroughly, and test responsibly within your authorized scope.

Key Takeaways

- 1. **Enumeration is Key:** Without a comprehensive understanding of the system, privilege escalation is unlikely to succeed. Always enumerate thoroughly before attempting exploitation.
- 2. **Prioritize Misconfigurations:** Look for weak permissions, exposed credentials, and overly permissive services. These are often the easiest entry points.
- 3. **Patch Awareness:** Outdated systems with missing patches are often prime targets. Use the `wmic qfe` command to understand patch history.
- 4. **System Logs:** Be mindful of leaving traces during enumeration. Activities like creating new users or enabling services can leave a trail for defenders.